

Q-FinOpt System Architecture & Technical Report

AI-Driven Quantitative Mutual Fund Prediction, Risk Modeling & Systematic Portfolio Advisory

EXECUTIVE SUMMARY

Q-FinOpt is an institutional-grade financial analytics platform designed for retail and semi-institutional mutual fund investors. It combines real-time AMFI NAV feeds, a 74.01% directional accuracy Multi-Factor Gradient Boosted ML model, 95% Confidence Corridor forecasting, Monte Carlo Systematic Withdrawal simulation, and a native Android client.

1. PROJECT BACKGROUND & PROBLEM SOLVED

Traditional retail mutual fund platforms present historical return percentages with static, retrospective charts. They provide zero predictive guidance on forward momentum, treat volatile drawdowns uniformly with upside volatility, and offer no quantitative optimization for exit or systematic withdrawal timing. Naive 1-day price direction models operate at a ~50% random-walk baseline (coin-flip zone).

Q-FinOpt bridges this gap by engineering a complete fintech pipeline: (1) 21-day forward multi-factor machine learning that captures intermediate trend momentum and risk characteristics, (2) dynamic volatility-calibrated path prediction with 95% confidence bounds (Bull/Bear corridors), (3) rigorous risk metrics (Sortino Ratio, Maximum Drawdown), and (4) Monte Carlo exit timing simulations based on 5,000 geometric Brownian motion trials.

2. END-TO-END TECHNOLOGY ARCHITECTURE

The platform is engineered across a modern, low-latency four-tier architecture:

Layer	Technologies Used	Key Responsibilities
Mobile Frontend	Kotlin 2.0, Jetpack Compose, Material3, Coroutines	Hardware-accelerated dynamic canvas charting, live conviction badges, SIP/SW
Backend API	FastAPI (Python 3.12), Uvicorn, Pydantic v2, Gunicorn	REST endpoints, async cache warming, CORS middleware, PDF report genera
Machine Learning	Scikit-Learn (HistGradientBoost), NumPy, SciPy, Jupyter	Multi-factor feature engineering, 21-day forward return inference, 95% confiden
Data Feeds	AMFI India Daily NAV, mfapi.in API, YFinance (NIFTY)	13,217 live mutual fund NAVs, 419,188 historical rows across 296 tracked sche

3. MACHINE LEARNING MODEL & EMPIRICAL BENCHMARKS

The core AI forecasting engine employs a Histogram-Based Gradient Boosted Decision Tree (HistGradientBoostingRegressor). The dataset contains 394,548 historical trading entries partitioned with a strict chronological 80/20 train/test time split (315,638 training instances up to cutoff, 78,910 out-of-time test instances).

Feature Engineering Matrix (12 Quantitative Factors):

- Multi-Horizon Momentum: 5-day, 10-day, and 21-day percentage returns (ret_5d, ret_10d, ret_21d)
- Volatility Dynamics: 10-day and 30-day rolling return standard deviations (vol_10d, vol_30d)
- Momentum Efficiency Ratio: $\text{ret_5d} / \text{vol_10d}$ (penalizes noisy momentum spikes)
- Oscillators & Normalized Position: 14-day RSI (momentum exhaustion) and 60-day NAV Z-score
- Cross-Sectional Risk Ratios: Fund Sharpe ratio, Alpha, Market Beta, and Expense Ratio

Performance Metric	Naive 1-Day Baseline	21-Day Momentum	Q-FinOpt Upgraded AI	Gain vs Baseline
Directional Accuracy	51.84% (Coin-Flip)	62.66%	74.01%	+22.17%
Precision	51.20%	75.27%	74.05%	+22.85%
Recall	50.90%	74.98%	99.90%	+49.00%
F1 Score	51.05%	75.12%	85.06%	+34.01%
High-Conviction (>Q3)	52.10%	75.04%	75.55%	+23.45%
Information Coeff (IC)	-0.000 (No signal)	+0.007	+0.0393 (p < 1e-10)	Statistically Significant

4. MATHEMATICAL FORMULATIONS & QUANTITATIVE FORMULAS

1. Multi-Factor Drift & Dynamic Forecast Path:

The model predicts the 21-day forward return y_{hat} . The daily drift is extracted via compound formula:

$$\mu_{ML} = (1 + y_{hat})^{(1/21)} - 1$$

The initial drift is blended: $\mu_{initial} = 0.70 * \mu_{ML} + 0.30 * \mu_{momentum}$. At each future trading step i:

$$drift(i) = \mu_{eq} + (0.94^i) * (\mu_{initial} - \mu_{eq}) + 0.25 * \sigma_{daily} * \sin(2 * \pi * i / 21)$$

2. 95% Confidence Corridor Envelopes:

To provide probabilistic risk corridors rather than unrealistic deterministic lines, Q-FinOpt calculates:

Upper Corridor (Bull 95%): $NAV(i) + 1.96 * \sigma_{daily} * \sqrt{i} * 100$

Lower Corridor (Bear 5%): $NAV(i) - 1.96 * \sigma_{daily} * \sqrt{i} * 100$

3. Downside Risk Metric (Sortino Ratio):

Sortino isolates downside risk by penalizing only negative return deviation (downside semi-deviation):

$$Sortino = (\mu_{daily} * 252) / (\sigma_{downside} * \sqrt{252})$$

For HDFC Infrastructure Fund, while Sharpe is 1.300, Sortino reaches 3.682, proving substantial upside asymmetry.

4. Maximum Drawdown (MDD):

Measures peak-to-trough historical loss to calibrate downside tolerance: $MDD = \min_t [(NAV_t - Peak_t) / Peak_t]$.

Evaluated on HDFC Infrastructure Fund: -22.36%.

5. FUND ANALYTICS CASE STUDY: HDFC INFRASTRUCTURE FUND

Evaluated on the full historical dataset (Scheme Code: 101132):

Metric	Value	Interpretation & Significance
Current NAV	Rs 63.27	Official latest NAV valuation
Daily Mean Return	0.1522%	Historical day-over-day growth rate
Annualized Expected Return	38.35%	Compounded historical annualized return trajectory
Daily Volatility (SD)	1.1427% (Ann: 18.14%)	Calibrated market dispersion benchmark
Sharpe Ratio	1.300	Excess return per unit of total risk
Sortino Ratio	3.682	Superior return per unit of downside risk
Alpha	2.650	Outperformance generated above benchmark index
Beta	0.880	Market sensitivity (< 1.0 indicates dampened market volatility)
Maximum Drawdown (MDD)	-22.36%	Historical maximum correction drop
AI Signal & Conviction	ACCUMULATE (68.0%)	Multi-factor model recommendation for current regime

6. APPLICATION FEATURES & REST API SPECIFICATIONS

Mobile Screens & Capabilities:

- Analysis Screen: Dynamic actual vs 30-day forecast curves with 95% bull/bear corridors, AI signal badges, and statistical indicators.
- Compare Screen: Multi-fund relative percentage change comparison across configurable time horizons (1M to ALL).
- Systematic Withdrawal Plan (SWP): 5,000 Monte Carlo simulations projecting optimal withdrawal day and profit probability.
- SIP Wealth Planner: Year-by-year nominal and inflation-adjusted corpus accumulation with exact root-found XIRR.
- Live AMFI Search: Live search interface scanning 13,217 AMFI-registered mutual fund schemes with min/max statistics.

Production RESTful API Endpoints:

Method	Endpoint Route	Description & Response Payload
GET	/api/market	Live indices (NIFTY 50, Sensex) and market sentiment
GET	/api/funds/filter	Filter funds by category (Large, Mid, Flexi) and risk level
GET	/api/funds/{name}/stats	Quantitative indicators: Sharpe, Sortino, Alpha, Beta, MDD, Signal
GET	/api/funds/{name}/history	Daily NAV history, AI forecast, 95% corridors, return distribution

POST	/api/funds/compare	Multi-fund comparative normalized series and comparative stats table
POST	/api/simulate/withdrawal	Monte Carlo SWP simulation, P5/P95 bands, optimal exit day
POST	/api/simulate/sip	Systematic investment growth projection, XIRR, percentiles
GET	/api/nav/search	Real-time AMFI India live NAV database search across 13k+ funds
POST	/api/report/pdf	Generate and download branded PDF investment analysis report
GET	/download	Direct download endpoint for QFinOpt.apk mobile installer

7. VERIFICATION, PACKAGING & DEPLOYMENT

- Android Package: QFinOpt.apk (20.02 MB), assembled with Gradle 9.1 and Android API 34.
- Live Local Server: Hosted on host Wi-Fi interface (<http://10.0.56.180:8000>) with active CORS & background cache warmup.
- Source Control: Git repository fully synchronized at <https://github.com/uppalamanisai5-a11y/qfinopt-backend.git>.